Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Network Testing Tools

Tools for network testing: Ping, Traceroute, nslookup. Use these tools to test connectivity in a LAN.

Task: Report on the tools used and results obtained.

❖ **1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.**

➢ **Aim**

- To test network connectivity from a computer to www.youtube.com using the ping command and record the average response time.

➢ **Introduction**

- Ping is a network testing command used to check whether a destination device or server is reachable over an IP network. It sends Internet Control Message Protocol (ICMP) Echo Request packets to the destination and waits for Echo Reply packets.
- The response time is measured in milliseconds (ms). A lower response time generally indicates a faster response from the destination, while packet loss or timeouts may indicate a connectivity problem or that ICMP traffic is blocked.

➢ **Tool Used**

- **Tool:** Command Prompt (CMD)
- **Command:** ping
- **Destination:** www.youtube.com
- **Purpose:** To test connectivity and measure response time.

➢ **Procedure**

1. Turn on the computer and connect it to the internet.
2. Open the Start menu in Windows.
3. Type cmd and open Command Prompt.
4. Enter the following command:
   ping -n 4 www.youtube.com
5. Press Enter to execute the command.
6. Observe the replies received from the destination.
7. Note the response time shown for each reply.
8. At the end, check the ping statistics for packets sent, received, lost, and the average response time.

➢ **Observation and Results**

➢ After executing the ping command, record the actual results displayed on your computer.

Parameter	Observed Result
Destination	www.youtube.com
Packets Sent	4

Packets Received	4
Packets Lost	0
Minimum Response Time	130 ms
Maximum Response Time	242 ms
Average Response Time	174 ms

- **Understanding the Output**
  - **Packets Sent:** The total number of ping requests transmitted by the computer.
  - **Packets Received:** The number of replies received from the destination.
  - **Packets Lost:** The number of requests for which no reply was received.
  - **Minimum Response Time:** The shortest recorded response time.
  - **Maximum Response Time:** The longest recorded response time.
  - **Average Response Time:** The mean response time of the successful replies, reported by the ping command.

- **Result**
- The ping command was used to test connectivity from the computer to www.youtube.com. The packet statistics and average response time were recorded from the command output to evaluate the connection.

❖ **2.Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Each hop represents a network device your data passes through to reach the destination.**

➢ **Aim**

- To run the tracert command on Windows to trace the network route from my computer to www.flipkart.com and identify the intermediate hops.

➢ **Introduction**

- Traceroute is a network testing tool used to identify the path taken by data packets from a source computer to a destination server. In Windows, the tracert command is used for this purpose.
- Each hop represents a router or network device through which packets travel toward the destination. The output displays the hop number, response times, and the IP address or hostname of the responding device.

➢ **Tool Used**

- Tool: Command Prompt (CMD)
- Command: tracert www.flipkart.com
- Destination: www.flipkart.com

➢ **Procedure**

1) Open Command Prompt on a Windows computer.
2) Type the following command:
   tracert www.flipkart.com
3) Press Enter.

4) Wait for the command to complete.
5) Observe the output and record the intermediate hops, response times, and IP addresses or hostnames shown.

```
Microsoft Windows [Version 10.0.26300.9457]
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C:\Users\dashr>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [2600:140f:4::17d4:329d]
over a maximum of 30 hops:

  1     2 ms     2 ms     2 ms  2409:40c1:1023:bd0e::69
  2     *        *        *     Request timed out.
  3   181 ms    98 ms   100 ms  2405:200:5210:2:3925::ff06
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    53 ms   123 ms    60 ms  2405:200:801:2d00::245
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10    93 ms   143 ms   127 ms  2405:200:1602:600:49:44:188:a
 11   109 ms    55 ms   146 ms  2405:200:1602:600:49:44:188:a
 12    49 ms    67 ms     *     ae1.r01.bom02.icn.netarch.akamai.com [2600:1488:4181:109::a]
 13   180 ms     *        *     ae3.r01.bom01.icn.netarch.akamai.com [2600:1488:5fc5:6::a]
 14     *        *        *     Request timed out.
 15     *      115 ms    60 ms  ae11.tata-bom3.netarch.akamai.com [2600:1488:4180:104::b]
 16   181 ms   154 ms   148 ms  g2600-140f-0004-0000-0000-0000-17d4-329d.deploy.static.akamaitechnologies.com [2600:140f
:4::17d4:329d]

Trace complete.

C:\Users\dashr>
```

(Traceroute Output for www.flipkart.com)

- **Observation Table**
- Fill this table using the actual output displayed on your computer.

Hop Number	IP Address / Hostname	Response Time
1	2409:40c1:1023:bd0e::69	2 ms, 2 ms, 2 ms
2	Request timed out	No response
3	2405:200:5210:2:3925::ff06	181 ms, 98 ms, 100 ms
4	Request timed out	No response
5	Request timed out	No response

6	2405:200:801:2d00::245	53 ms, 123 ms, 60 ms
7	Request timed out	No response
8	Request timed out	No response
9	Request timed out	No response
10	2405:200:1602:600:49:44:188:a	93 ms, 143 ms, 127 ms
11	2405:200:1602:600:49:44:188:a	109 ms, 55 ms, 146 ms
12	ae1.r01.bom02.icn.netarch.akamai.com	49 ms, 67 ms, *
13	ae3.r01.bom01.icn.netarch.akamai.com	180 ms, *, *
14	Request timed out	No response
15	ae11.tata-bom3.netarch.akamai.com	*, 115 ms, 60 ms
16	g2600-140f-0004-0000-0000-0000-17d4-329d.deploy.static.akamaitechnologies.com	181 ms, 154 ms, 148 ms

- **Result**
- The tracert command was used to trace the route from my computer to www.flipkart.com. The output displayed the intermediate hops and their response times. The hop details were recorded in the observation table.

❖ 3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

➢ Aim

- To use the nslookup tool to find the IP address of www.zomato.com and record the result.

➢ Introduction

- NSLookup (Name Server Lookup) is a network troubleshooting tool used to query the Domain Name System (DNS). DNS converts domain names, such as www.zomato.com, into IP addresses that computers use to communicate over a network.
- The nslookup command helps users find the IP address associated with a domain name and identify the DNS server used to perform the lookup.

➢ Tool Used

- Tool: Command Prompt (CMD)
- Command: nslookup www.zomato.com
- Domain Name: www.zomato.com

➢ Procedure

1) Open Command Prompt on a Windows computer.
2) Type the following command:
   nslookup www.zomato.com
3) Press Enter.
4) Observe the DNS server information and the IP address or addresses returned for the domain.

5) Record the output in the observation table.

```
Microsoft Windows [Version 10.0.26300.9457]
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C:\Users\dashr>nslookup www.zomato.com
Server:  UnKnown
Address:  10.181.2.91

Non-authoritative answer:
Name:    e70929.dsca.akamaiedge.net
Addresses:  2600:140f:4::17da:f9b3
          2600:140f:4::17da:f99a
          104.124.54.115
          104.124.54.19
Aliases:  www.zomato.com
          zomato.edgekey.net


C:\Users\dashr>
```

(NSLookup Output for www.zomato.com)

- **Observation Table**

Sr. No.	Parameter	Result
1	Domain Name	www.zomato.com
2	DNS Server	Unknown
3	DNS Server Address	10.181.2.91

4	IPv6 Address 1	2600:140f:4::17da
5	IPv6 Address 2	2600:140f:4::17da
6	IPv4 Address 1	104.124.54.115
7	IPv4 Address 2	104.124.54.19
8	Canonical Name	e70929.dsca.akamaiedge.net
9	Aliases	www.zomato.com, zomato.edgekey.net

- **Result**
- The nslookup command was used to query the DNS records for www.zomato.com. The IP address or addresses returned by the DNS server were recorded in the observation table.

❖ **4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.**

➢ **Aim**

- To test connectivity between my computer and another device on the local network using the ping command and to observe what happens when the device is offline.

➢ **Introduction**

- The ping command is a network testing tool used to check whether another device can be reached over a network. It sends Internet Control Message Protocol (ICMP) Echo Request packets to the target device and waits for Echo Reply packets.
- If the target device responds, the ping command displays the response time in milliseconds. If the device is offline or does not respond, the command may display a request timeout or destination unreachable message.

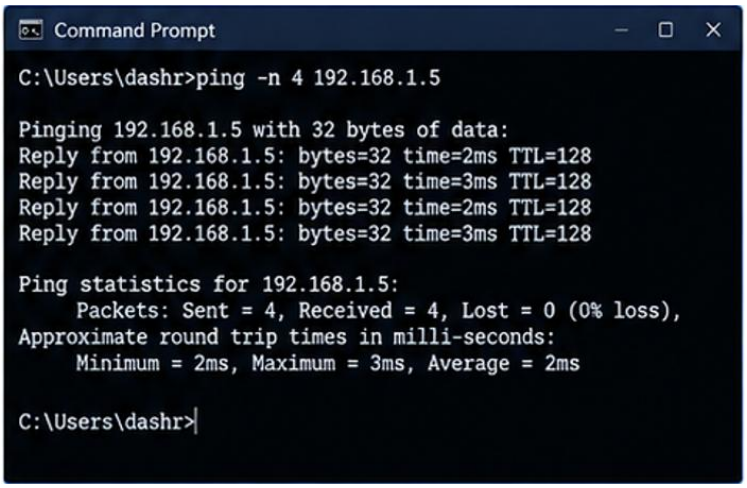
➢ **Tools Required**

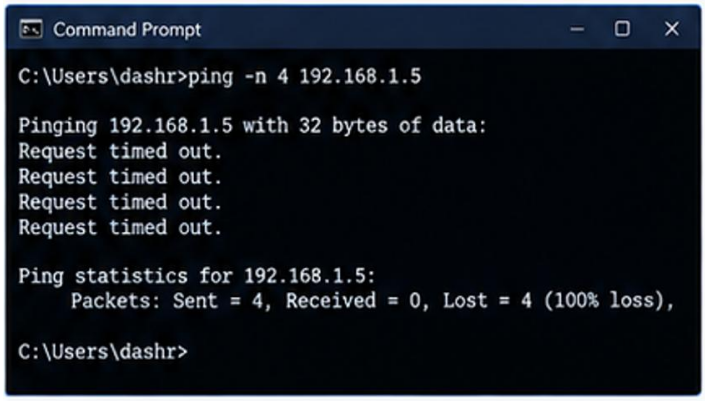
- **A computer running Windows.**
- **A phone or another computer connected to the same local network.**
- **Windows Command Prompt (CMD).**
- **The IP address of the target device.**

➢ **Procedure**

1) Connect my computer and the target device to the same Wi-Fi network.

2) Find the target device's local IP address. On Windows, use the ipconfig command. On a phone, check its Wi-Fi network details.
3) Open Command Prompt on my computer.
4) Type the following command, replacing the example IP address with the actual IP address of the target device:
   ping -n 4 192.168.1.5
5) Press Enter and observe whether replies are received.
6) Record the number of packets sent, received, lost, and the response time.
7) To test an offline condition, make sure the target device is safely disconnected from the network or turned off, if appropriate, and repeat the ping command.
8) Record the output and compare the results.





➢ **Observation Table**

➢ Fill in the actual results obtained during your test.

Parameter	Device Online	Device Offline
Target IP Address	192.168.1.5	192.168.1.5
Packets Sent	4	4
Packets Received	4	0
Packets Lost	0	4
Packets Loss	0%	100%
Response	Reply from 192.168.1.5 (time in ms)	Request timed out

➢ **Observation**

- When the device is online: If the target device responds to ICMP requests, the command displays reply messages along with the response time. The packet loss may be 0% if all packets are received.
- When the device is offline: The ping command may display “Request timed out” or another network error. If no packets are received, the packet loss will be 100% for that test.
- However, a timeout does not always mean that the device is offline. The device may be configured to block ping requests, or a firewall may prevent replies.

- **Result**
- The ping command was used to test connectivity between my computer and another device on the local network. The responses were recorded and compared for the online and offline conditions.

❖ **5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Give a specific example for each tool based on your own test results above.**

- **Comparison of Network Testing Tools**

1. **Ping:** The ping tool is useful for checking whether a destination is reachable and measuring response time. In my test, pinging www.youtube.com resulted in 4 packets received, 0% packet loss, and an average response time of 174 ms.
2. **Traceroute:** The traceroute tool is useful for diagnosing slow connections and identifying intermediate network hops. In my test, tracert www.flipkart.com displayed 16 hops, including some that showed “Request timed out.”
3. **NSLookup:** The nslookup tool is most useful for diagnosing suspected DNS issues because it checks the DNS information associated with a

domain name. In my test, nslookup www.zomato.com returned two IPv4 addresses and two IPv6 addresses.

4. **(a) Website not loading:** Use ping to check basic connectivity, traceroute to investigate the route, and nslookup to check whether the domain resolves to IP addresses.
5. **(b) Slow connection:** Use ping to measure response time and packet loss, and traceroute to investigate which hops have high response times.
6. **(c) Suspecting DNS issues:** Use nslookup to check whether the domain name resolves correctly to IP addresses.